

Incident Report: INC-2025 (P-204 Seal Failure)

Document Type: Incident Investigation Report

Incident ID: INC-2025

Date of Occurrence: 15-May-2025

Equipment Involved: Reflux Pump P-204, Pump P-205

Unit: Crude Distillation Unit (CDU)

INCIDENT DESCRIPTION:

At 14:30 hrs, a significant hydrocarbon leak was detected from the mechanical seal of pump P-204. The unit was safely shut down. A subsequent check of the standby pump, P-205, revealed that its coupling guard had been removed during maintenance under WO-2025-0441 and never reinstalled.

ROOT CAUSE ANALYSIS:

1. The mechanical seal on P-204 failed due to the use of an incompatible gasket. SOP-17 incorrectly specifies Graphite Grade A gaskets, whereas the fluid service requires Graphite Grade C.
2. The missing coupling guard on P-205 is a direct violation of Factory Act Section 41.

RECOMMENDATIONS:

1. Immediately revise SOP-17 to specify Graphite Grade C gaskets. Review all other pump SOPs for similar errors.
2. Audit all rotating equipment in the CDU to ensure coupling guards are installed.
3. Deploy a proactive warning to all field technicians regarding proper gasket selection and guard installation.